

# SALAMI

## 3 Independent Layers of Labels:

### 1) Musical Similarity (A, B, C, ... Z)

- essentially the different sections like a chorus, verse
- Z: special notation for a section v. diff. from rest of song
- ' : prime symbol for similar but slightly diff. prev. section
- A/a: large vs. small scale sections like sections vs. sub-sections

### Notes:

- Each uppercase letter must have 1+ lowercase letter
- Lower case letters are unique throughout the piece:  
"a" under A means the same musical idea as "a" under F.

### 2) Musical Function

- Verse
- Chorus
- Bridge
- Intro
- Outro
- Solo
- Other (fine distinctions between these major labels)

### 3) Instrumentation

- type of instrument leading the melody (includes vocals)

- possible for multiple instruments or even none

## MSAF

### Labelling Algorithms:

1) Structural Segmentation of Musical Audio via Constrained Clustering

- 2008

- Uses constrained clustering to improve methods at the time

2) SIPLCA analysis of chroma features using

harmonic pattern detection

- 2011

- Uses both but unclear exactly how.

- Only has:

- start time

- end time

- segment # (musical similarity label of section)

### Datasets:

- Human-annotated

- Uses other open-sourced datasets like SALAMI

1) Sagon: 30-min of heavy metal

2) SPAM: 5 human annotations for 50 tracks

- duration of section
- confidence
- value (musical similarity of section)
- time start of section

3) The Beatles: only Beatles

## Evaluating Hierarchical Structure in Music Annotations

Compare SALAMI and SPAM annotator agreement amounts

## Feature Engineering Paper

### Method:

- 1) Labels of music are refactored into 9 types and refined to each beat
- 2) A beat-wise feature extraction is studied to segment music based on beat
- 3) A Resnet-34 using self-attentiveness mechanism predicts the categories of each beat
- 4) Post-Processing to filter predicted labels

Notes: - Uses SALAMI

- 3% more optimal than current HR3F method
- They only use Annotator #1 to reduce ambiguity

## Proposed Label Design

1) Section Boundaries (time start, time end)

2) Section Similarity Label (A, B, ...)

↳ Need to ensure same labelling method as SALAMI for SPAM

3) Annotator ID

4) Musical Function (Verse, Chorus, ...)

↳ Optional since only SALAMI has it

## Dropped:

- Instrumentation
- Confidence

## Annotator Handling:

1) Single Annotator

- Easy
- Potential bias/inaccuracy

## 2) Soft-Label Distributions

- More Work
- Captures uncertainty using loss-weighting design
- Backed by the paper evaluating SALAM + SPAM:  
"Annotator disagreements reflect different valid hierarchical interpretations, not noise"

Essentially: both right

1) Verse A - Chorus - Verse B

2) Verse A (1) - Verse A (2) - Chorus - Verse B  
divided into 2

- Assign "confidence" values to SALAM (not same as SPAM self-attributed "confidence") and to SPAM  
↳ could use the self-attributed confidence to SPAM too tho